

NONCLINICAL REPORT

Nonclinical Report No. UTMB01

Evaluation of Immunogenicity and Efficacy of mRNA-1273 in Golden Syrian Hamster Model

SPONSOR:

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

PERFORMING LABORATORY:

University of Texas Medical Branch
301 University Blvd
Galveston, TX 77555

DATE

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LIST OF ABBREVIATIONS

Abbreviation	Definition
CoV	coronavirus
COVID-19	coronavirus disease 2019
DSPC	1,2-distearoyl-sn-glycero-3-phosphocholine and heptadecan-9-yl
ELISA	enzyme-linked immunosorbent assay
FBS	fetal bovine serum
IgG	immunoglobulin G
IHC	immunohistochemistry
IM	intramuscular(ly)
LNP	lipid nanoparticle
LOD	limit of detection
MEM	Minimum Essential Medium
mRNA	messenger RNA
NP	nucleoprotein
OD	optical density
PBS	phosphate-buffered saline
PEG	polyethylene glycol
PEG2000-DMG	1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol
PFU	plaque-forming unit(s)
PRNT	plaque reduction neutralization test
qRT-PCR	quantitative real-time polymerase chain reaction
RBD	receptor binding domain
S	spike
S-2P	spike protein with 2 proline substitutions within the heptad repeat 1 domain
sgRNA	subgenomic RNA
SARS-CoV-2	2019 novel coronavirus
SM-102	heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate
UTMB	University of Texas Medical Branch
UTR	untranslated region

QUALITY ASSURANCE STATEMENT

The study was not conducted according to Good Laboratory Practice regulations.

1. SUMMARY

Nonclinical studies have demonstrated that coronavirus (CoV) spike (S) proteins are immunogenic and that 2019 novel coronavirus (SARS-CoV-2) prefusion-stabilized S protein with 2 proline substitutions within the heptad repeat 1 domain (S-2P)-based vaccines, including those based on messenger RNA (mRNA) delivery platforms, are protective in animals. In addition, previous clinical studies of vaccines targeting related CoVs and other viruses have demonstrated that mRNA-based vaccines are safe and immunogenic.

The objectives of Study UTMB01 were to evaluate the immunogenicity of mRNA-1273 and assess the level of protection it provides from SARS-CoV-2 challenge in golden Syrian hamsters.

ModernaTX, Inc. (Sponsor) is developing mRNA-1273, a novel lipid nanoparticle (LNP) encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 contains a single mRNA that encodes for the SARS-CoV-2 S glycoprotein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P). The mRNA-1273 Drug Product consists of the mRNA formulated in a mixture of 4 ionizable, structural, helper, and polyethylene glycol (PEG) lipids common to the Sponsor's mRNA vaccine platform: heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate (SM-102); cholesterol; 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC); and 1-monomethoxypolyethyleneglycol-2,3-dimyristylglycerol polyethylene glycol (PEG2000-DMG), respectively. The test article in this study was mRNA-1273 Drug Product (WH2020_NatSP_2P, Tech dev DP). Phosphate-buffered saline (PBS) was used as a negative control.

Female Syrian hamsters, 6 to 7 weeks old (n = 15/group), were administered 1, 5, or 25 µg of mRNA-1273 on a prime (Week 0, Day 1)/boost (Week 3, Day 22) schedule or 25 µg of mRNA-1273 or PBS administered on a prime-only (Week 0, Day 1) schedule. An equal volume of the vaccine was administered by intramuscular (IM) injection into each hind leg (50 µL per dose site) at each dose administration. At Week 3 (Day 21; 1 day prior to the boost dose) and Week 6 (Day 42), sera were collected to assess SARS-CoV-2 S- and receptor binding domain (RBD) specific immunoglobulin G (IgG) antibody titers by enzyme linked immunosorbent assay (ELISA) and neutralizing activity by live virus plaque reduction neutralization test (PRNT).

At Week 6 (Day 43), all animals were challenged intranasally with 10⁵ plaque-forming units (PFU) of wild-type SARS-CoV-2 (USA-WA1/2020). Body weight was measured through Day 14 post-challenge. Sera were collected at Day 2, Day 4, and Day 14 post-challenge to assess S- and nucleoprotein (NP)-specific IgG antibody titers by ELISA as well as neutralizing activity by live virus PRNT. Animals were euthanized, and lungs and nasal turbinates were collected on Day 2 (n = 5/group), Day 4 (n = 5/group), and Day 14 (n = 5/group) post-challenge to evaluate viral load (live virus by plaque assay), actively replicating virus (subgenomic RNA [sgRNA] by quantitative

real-time polymerase chain reaction [qRT-PCR]), and lung histopathology (including immunohistochemistry [IHC]).

mRNA-1273 is immunogenic in Syrian hamsters. Immunization with mRNA-1273 elicited S- and RBD-specific IgG antibody titers in all groups and neutralizing antibody responses in all animals in the prime/boost groups and a subset of animals in the prime-only group. In general, the post-boost S- and RBD-specific IgG antibody titers and neutralizing antibody titers were higher than post prime titers in animals immunized with 1, 5, and 25 µg of mRNA-1273 on a prime/boost schedule. Complete neutralizing seroconversion in all animals was only achieved after administration of the boost dose. Post-boost (Day 42) RBD-specific IgG antibody titers and neutralizing antibody titers were also higher in animals in the prime/boost groups than titers at Day 42 in animals immunized with a prime-only dose of 25 µg of mRNA-1273. The post-boost neutralizing antibody titers in animals immunized with 1, 5, and 25 µg of mRNA-1273 on a post/boost schedule met or exceeded the highest titers observed in convalescent sera.

After challenge with wild-type SARS-CoV-2, minimal weight loss was observed in animals immunized with a prime/boost dose of 1, 5, and 25 µg of mRNA-1273 through Day 14 post-challenge, except for one hamster in the 5 µg mRNA-1273 group. Excluding the animals necropsied on Day 2 and Day 4 post-challenge, the mean maximum weight loss in the mRNA-1273 prime/boost groups was 2.25% over 14 days post-challenge. All but one animal immunized with a prime-only dose of 25 µg of mRNA-1273 experienced little or no weight loss; the mean maximum weight loss in this group was 6.2%. In contrast, animals in the PBS group experienced marked weight loss by Day 6 post-challenge; the mean weight increased from Day 7 to Day 14 post-challenge but remained lower in the PBS group than in all other groups. The mean maximum weight loss in the PBS group was 12%.

In the PBS group, S- and NP-specific IgG antibody titers and neutralizing antibody titers were detected at Day 14 post-challenge, and the levels were significantly higher than the titers at Day 42 pre-challenge. In animals immunized with mRNA-1273 (all groups), S-specific IgG antibody titers and neutralizing antibody titers were detected prior to the virus challenge and the titers increased significantly by Day 14 post-challenge, except for neutralizing antibody titers in animals immunized with a prime/boost dose of 1 µg of mRNA-1273, in which the difference was not significant. In addition, NP-specific antibody titers were detected in all groups at Day 14 post-challenge, although the titers were lower in the mRNA-1273 groups than in the PBS group. These results indicate that although mRNA-1273 protects Syrian hamsters from weight loss after SARS-CoV-2 challenge, this protection is likely not fully sterilizing as vaccinated animals mounted an anamnestic immune response against the virus challenge.

No protection from SARS-CoV-2 was observed in animals administered PBS, as substantial viral load (viral titers by plaque assay) and viral replication (sgRNA by qRT-PCR) were observed in

the lungs and nasal turbinates on Day 2 and Day 4 post-challenge. Measurement of viral load showed that animals immunized with 1, 5, and 25 µg of mRNA-1273 on a prime/boost schedule were protected from virus challenge, with mean viral titers that were below the limit of detection (LOD) in the lungs and nasal turbinates at Day 4 post-challenge. In addition, all animals administered a prime/boost dose of 5 and 25 µg of mRNA-1273 and all but one animal administered a prime/boost dose of 1 µg of mRNA-1273 had no detectable viral replication (sgRNA) in the lungs by Day 4 post-challenge. In the nasal turbinates, all animals in the prime/boost groups were partially protected, as the viral replication (sgRNA) observed on Day 2 and Day 4 post-challenge was significantly lower than that in the PBS control group. A prime-only dose of 25 µg of mRNA-1273 partially protected animals, as live virus and viral replication were observed on Day 2 post-challenge in lungs and nasal turbinates; however, the mean viral titers were lower than those in the PBS group, and by Day 4 post-challenge, the mean viral titers were below the LOD.

Lungs of hamsters evaluated histologically after challenge with wild-type SARS-CoV-2 showed that over the 14-day infection course, animals immunized with a prime/boost dose of mRNA-1273 generally displayed mild to moderate inflammation with viral antigens that were either absent or minimally abundant at Day 4 post-challenge. Lung sections of animals immunized with a prime-only dose of 25 µg of mRNA-1273 exhibited inflammation that was largely associated with perivascular and peribronchiolar regions in both a focally diffuse or multifocal distribution, but this inflammation was generally milder than that in animals in the PBS control group at the same time point and did not persist through Day 14. One outlier animal in each of the 1 µg prime/boost and 25 µg prime-only groups exhibited more severe histopathological phenotypes, with the highest levels of viral antigen detected among all vaccinated hamsters. These 2 animals did not exhibit drastic weight loss and the animal in the 25 µg prime-only group had live virus and virus replication in the lungs and nasal turbinates that was similar to that in other animals in their dose group.

Overall, mRNA-1273 administered on a prime/boost dosing schedule is immunogenic and efficacious and provides protection from SARS-CoV-2 in Syrian hamsters.

2. DESCRIPTION OF TEST PROCEDURES

2.1. Conduct of Study

2.1.1. Sponsor

ModernaTX, Inc.
200 Technology Square
Cambridge, MA 02139

2.1.2. Performing Laboratory

University of Texas Medical Branch (UTMB)
301 University Blvd
Galveston, TX 77555

2.1.3. Study Number

Study number UTMB01

2.1.4. Purpose of the Study

The objectives of this study were to evaluate the immunogenicity of mRNA-1273 and assess the level of protection it provides from SARS-CoV-2 challenge in golden Syrian hamsters.

2.1.5. Institutional Animal Care and Use Committee Approval

All animal experiments described in this study report were approved and conducted in accordance with the regulations of the UTMB Institutional Animal Care and Use Committee (IACUC Protocol 2003039).

2.1.6. Study Director

(b) (6)

UTMB
301 University Blvd
Galveston, TX 77555

2.1.7. Key Personnel

Role	Study UTMB01
Study coordinator	(b) (6) UTMB 301 University Blvd Galveston, TX 77555
Sample processing in high containment	(b) (6) UTMB 301 University Blvd Galveston, TX 77555
High containment program coordinator of animal studies	(b) (6) UTMB 301 University Blvd Galveston, TX 77555
Veterinary pathologist	(b) (6) UTMB 301 University Blvd Galveston, TX 77555 (b) (6) National Institutes of Health 12735 Twinbrook Parkway, Room 2W-01 Rockville, MD 20852-1770
Report compilation	PPD 3900 Paramount Parkway Morrisville, NC 27560

2.1.8. Dates of Study Study Performance

Schedule of Events	Dates
Test article administration	
Prime dose (prime/boost groups)	28 April 2020 (Week 0, Study Day 1)
Prime dose (prime-only and PBS groups)	01 May 2020 (Week 0, Study Day 1)
Boost dose (prime/boost groups)	19 May 2020 (Week 3, Study Day 22)
Serum collection (S- and RBD-specific IgG antibody, neutralizing activity)	
Prime/boost groups	18 May 2020 (Week 3, Study Day 21); 08 June 2020 (Week 6, Study Day 42)
Prime-only and PBS group	21 May 2020 (Week 3, Study Day 21); 11 June 2020 (Week 6, Study Day 42)
Challenge with SARS-CoV-2	
Prime/boost groups	09 June 2020 (Week 6, Study Day 43)
Prime-only and PBS group	12 June 2020 (Week 6, Study Day 43)
Lung and nasal turbinate collection	
Serum collection (post-challenge S- and NP-specific IgG antibody responses and neutralization activity; all groups)	Day 2, Day 4, and Day 14 after SARS-CoV-2 challenge

2.1.9. Records Maintained

Data is stored at UTMB in electronic format, which is password protected and can be accessed only by authorized personnel. The server is maintained by the UTMB information technology staff and is backed up remotely on a daily basis.

2.2. Test Articles

The test article in this study was mRNA-1273. Phosphate-buffered saline was used as a negative control.

2.2.1. mRNA-1273

mRNA-1273 contains a single mRNA that encodes SARS-CoV-2 S-2P formulated into a mixture of 4 lipids: SM-102, cholesterol, DSPC, and PEG2000-DMG. A summary of the dose formulation analysis results is presented in [Table 1](#).

The mRNA-1273 Drug Product used in this study was an mRNA formulation (WH2020_NatSP_2P, Tech dev DP, AMPDP-200005) prepared with the same method as the Good Manufacturing Practice mRNA-1273 clinical Drug Product ([Appendix 1](#)). This product is a

novel LNP-encapsulated mRNA-based vaccine against SARS-CoV-2. mRNA-1273 encodes for the full-length S protein of SARS-CoV-2, modified to introduce 2 proline residues to stabilize the S protein into a prefusion conformation (S-2P). mRNA-1273 consists of mRNA encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process as described previously ([Hassett et al 2019](#)).

Preclinical mRNA-1273 vaccine was manufactured by the Sponsor. A sequence-optimized mRNA encoding prefusion-stabilized SARS-CoV-2 S-2P protein was synthesized in vitro using an optimized T7 RNA polymerase-mediated transcription reaction with complete replacement of uridine by N¹m-pseudouridine ([Nelson et al 2020](#)). The reaction included a DNA template containing the immunogen open-reading frame flanked by 5' untranslated region (UTR) and 3' UTR sequences and was terminated by an encoded polyA tail. After transcription, the Cap 1 structure was added to the 5' end using vaccinia capping enzyme (New England Biolabs) and vaccinia 2'O-methyltransferase (New England Biolabs). The mRNA was purified (b) (4), buffer exchanged (b) (4), sterile filtered, and kept frozen at –20°C until further use.

mRNA-1273 was encapsulated in an LNP through a modified ethanol-drop nanoprecipitation process as previously described ([Hassett et al 2019](#)). (b) (4)

The mixture was neutralized with tris(hydroxymethyl)aminomethane hydrochloride (Tris HCl), pH 7.5, sucrose was added as a cryoprotectant, and the final solution was sterile filtered. Vials were filled with mRNA-1273-formulated LNP and stored frozen at –70°C until further use. The Drug Product underwent analytical characterization, which included determinations of particle size and polydispersity, encapsulation, mRNA purity, double-stranded RNA content, osmolality, pH, endotoxin, and bioburden, and the material was deemed acceptable for the in vivo study ([Corbett et al 2020](#)).

Table 1: Test Materials and Summary of Dose Formulation Analysis Results

Test Material	mRNA Description	LNP Lot No.	mRNA Lot No.	Size (nm)	PDI	EE%
mRNA-1273 Drug Product	mRNA SARS-CoV-2 spike protein ^a	(b) (4)				

Abbreviations: EE% = encapsulation efficiency; LNP = lipid nanoparticle; mRNA = messenger RNA; PDI = polydispersity index; SARS-CoV-2 = 2019 novel coronavirus; SM-102 = heptadecan-9-yl 8-((2-hydroxyethyl)(6-oxo 6-(undecyloxy)hexyl)amino)octanoate.

Note: All mRNA test material stocks were formulated in SM-102-containing LNPs with a final storage buffer of 10.7 mM sodium acetate, 8.7% sucrose, and 20 mM tris(hydroxymethyl)aminomethane hydrochloride at pH 7.5 and stored frozen at -70°C.

The EE% is the percentage of mRNA (b) (4)

(b) (4)

^a Full-length SARS-CoV-2 spike protein stabilized in the prefusion conformation with 2 proline substitutions within the heptad repeat 1 domain (S-2P) and including a transmembrane domain for insertion into the cell membrane after translation.

^b Manufactured by ModernaTX, Inc. (01 Feb 2020).

2.2.2. Phosphate-Buffered Saline

Phosphate-buffered saline (b) (4) was purchased commercially.

2.3. Test System

Female golden Syrian hamsters, 6 to 7 weeks old, were obtained from Envigo (Indianapolis, Indiana).

Human convalescent serum was obtained from adults with mild, moderate, or severe CoV disease 2019 (COVID-19) and a history of laboratory-confirmed SARS-CoV-2 infection. Details are provided in [Section 2.4.4](#).

2.4. Methods

2.4.1. Study Design and Treatment Groups

Animals were treated according to the study design presented in [Table 2](#). Female Syrian hamsters, (n = 15/group) were administered 1, 5, or 25 µg of mRNA-1273 on a prime (Week 0, Day 1)/boost (Week 3, Day 22) schedule or 25 µg of mRNA-1273 or PBS administered on a prime-only (Week 0, Day 1) schedule. An equal volume of the vaccine was administered by IM injection into each hind leg (50 µL per dose site) at each dose administration. At Day 21 (1 day prior to boost dose) and Day 42, sera were collected to assess SARS-CoV-2 S- and RBD-specific IgG antibody titers by ELISA as well as neutralizing activity by live virus PRNT.

At Week 6 (Day 43), all animals were challenged intranasally with 10⁵ PFU of wild-type SARS-CoV-2 (USA-WA1/2020; GenBank: MN985325.1). This virus was a passage 5 stock,

expanded in Vero E6 cells from a passage 3 stock of SARS-CoV-2 USA_WA1/2020 obtained from the World Reference Center for Emerging Viruses and Arboviruses at UTMB. It is a SARS-CoV-2 isolate obtained from the first US patient in Washington identified on 22 January 2020. Body weight was measured through Day 14 post-challenge. Sera were collected at Day 2, Day 4, and Day 14 post-challenge to assess S- and NP-specific IgG antibody titers by ELISA as well as neutralizing activity by live virus PRNT. Animals were euthanized, and lungs and nasal turbinates were collected on Day 2 (n = 5/group), Day 4 (n = 5/group), and Day 14 (n = 5/group) post-challenge to evaluate viral load (live virus by plaque assay), actively replicating virus (sgRNA by qRT-PCR), and lung histopathology (including IHC).

Table 2: Treatment Regimen for Study UTMB01

Group (n = 15)	Treatment	Dose (µg)	Schedule			ROA (Volume)	Sample Collection	Experimental Assessments
			Prime	Boost	Challenge			
1	PBS (prime-only)	N/A	Week 0/ Day 1	NA	Week 6/ Day 43	IM (50 µL in each hind leg)	Sera (Week 3/Day 21, Week 6/Day 42)	S- and RBD-specific IgG antibody titers by ELISA and neutralization activity by live virus PRNT assay
2	mRNA-1273 (prime-only)	25	Week 0/ Day 1	NA				
3	mRNA-1273 (prime/boost)	25	Week 0/ Day 1	Week 3/ Day 22			Sera (Day 2, Day 4, and Day 14 post-challenge)	Weight through Day 14 post-challenge
4	mRNA-1273 (prime/boost)	5	Week 0/ Day 1	Week 3/ Day 22				S- and NP-specific IgG antibody titers by ELISA and neutralization activity by live virus PRNT assay
5	mRNA-1273 (prime/boost)	1	Week 0/ Day 1	Week 3/ Day 22			Lungs and nasal turbinates (Day 2, Day 4, and Day 14 post-challenge)	Viral load by plaque assay, viral replication (sgRNA by qRT-PCR), and lung histopathology

Abbreviations: ELISA = enzyme-linked immunosorbent assay; IgG = immunoglobulin G; IM = intramuscular; NA = not applicable; NP = nucleoprotein; PBS = phosphate-buffered saline; qRT-PCR = quantitative real-time polymerase chain reaction; PRNT = plaque reduction neutralization test; RBD = receptor binding domain; ROA = route of administration; S = spike; sgRNA = subgenomic RNA.

2.4.2. Enzyme-Linked Immunosorbent Assay

The 96-well microtiter plates ((b) (4)) were coated with 1 µg/mL SARS-CoV-2 S protein (Sino Biological #40589-V08B1), 1 µg/mL RBD subdomain antigen (Sino Biological #40592-V08H0) or 1 µg/mL SARS-CoV-2 NP (Sino Biological, #40588-V08B). After overnight

incubation at 4°C, plates were washed 4 times with PBS with 0.05% Tween-20 and blocked for 2 hours at 37°C (b) (4). After washing, 5-fold serial dilutions of hamster serum were added (assay diluent: PBS + 0.05% Tween-20 + 5% goat serum). Plates were incubated for 2 hours at 37°C and washed, and then horseradish peroxidase conjugated goat anti-Syrian hamster IgG (Abcam AB7146) was added at 1:10,000 dilution in assay diluent. Plates were incubated for 1 hour at 37°C and washed, and then tetramethylbenzidine substrate (b) (4) (b) (4) was used to detect bound antibody. After incubation at room temperature for 10 minutes, the reaction was stopped by adding tetramethylbenzidine stop solution (b) (4) (b) (4), and the absorbance was measured at optical density (OD) 450 nm. Titers were determined using a 4-parameter logistic curve fit in GraphPad Prism (GraphPad Software, Inc.) and defined as the reciprocal dilution at OD_{450nm} = 1.0 (normalized to a hamster standard on each plate).

2.4.3. Plaque Reduction Neutralization Test

Two-fold serial dilutions of heat-inactivated serum at an initial dilution of 1:10 were prepared in serum-free Minimum Essential Medium (b) (4) and incubated with mNeonGreen SARS-CoV-2 for 1 hour at 37°C at a final concentration of 100 PFU (Xie et al 2020). Virus-serum mixtures then were adsorbed onto Vero-E6 monolayers in 96-well plates for 1 hour at 37°C and replaced with MEM/methylcellulose/2% fetal bovine serum (FBS) overlay. After 2 days of incubation at 37°C in humidified 5% carbon dioxide, neon-green fluorescent protein plaques were visualized and counted. Neutralizing antibody titers were determined with an endpoint of 60% plaque reduction and compared with the titers of convalescent sera (2 high-, 2 medium-, and 2 low-titer subjects).

2.4.4. Human Convalescent-Phase Serum

Human convalescent sera (n = 6) were obtained from adults between 18 and 55 years old with mild (n = 2), medium (n = 2), and severe (n = 2) COVID-19 and a history of laboratory-confirmed SARS-CoV-2 infection 1 to 2 months before providing specimens. These specimens were obtained from Aalto Bio Reagents Ltd.

2.4.5. Weight Measurement

Body weight was measured from the day of SARS-CoV-2 challenge through Day 14 post-challenge.

2.4.6. Assessment of Viral Load in Lungs and Nasal Turbinates

The right lung and nasal turbinates were harvested from animals at Day 2 (n = 5 /group), Day 4 (n = 5 /group), and Day 14 (n = 5/group) post-challenge and homogenized in 700 µL of Leibovitz L-15 medium (b) (4) /10% FBS/1× Antibiotic-Antimycotic

(b) (4)) using the (b) (4) bead mill (b) (4) and 5-mm stainless steel beads (b) (4)). Viral loads (live virus) in tissues were measured by plaque assay. Briefly, 10-fold serial dilutions of lung and nasal turbinate homogenates were prepared in serum-free MEM media and adsorbed on Vero-E6 monolayers in a 48-well plate for 1 hour at 37°C. The virus inoculum was removed and replaced with an overlay of MEM/methylcellulose/2% FBS. After 3 days, plates were fixed with neutral buffered formalin. Plaques were immunostained with a human monoclonal antibody cocktail specific for the S protein (b) (4) and an anti-human IgG horseradish peroxidase-conjugated secondary antibody (b) (4)). Plaques were counted, and virus load per gram tissue was determined.

2.4.7. Quantification of SARS-CoV-2 RNA in Lungs and Nasal Turbinates

Replicating viral RNA was assessed in the lungs and nasal turbinates by measuring subgenomic SARS-CoV-2 E gene RNA by qRT-PCR using previously described primers, probe, and cycle conditions (Wölfel et al 2020). Briefly, RNA was extracted from lung and nasal turbinate homogenates using TRIZOL LS (b) (4)) and a Direct-zol RNA Microprep kit (b) (4)). Then, 0.5 µg of RNA was reverse transcribed using Superscript IV (b) (4)) according to the manufacturer's recommendations. Quantitative real-time polymerase chain reaction was performed using (b) (4) Fast Advanced Master Mix (b) (4)) and primers and a FAM-ZEN/Iowa Black FQ labeled probe sequence (IDT) on the QuantStudio 6 system (Applied Biosystems). An Ultramer DNA oligo (IDT) spanning the amplicon was used to generate standard curves to calculate the sgRNA copies per gram of tissue.

2.4.8. Histology

Lung samples were processed per a standard protocol for histological analysis. Briefly, lower left lung lobe tissue samples were placed in 10% neutral buffered formalin until adequately fixed. Thereafter, tissues were placed into cassettes and embedded in paraffin. Paraffin tissue blocks were subsequently sectioned (5 µm) and stained with hematoxylin and eosin. Sections were examined by a board-certified veterinary pathologist using an (b) (4) light microscope. An (b) (4) (b) (4) camera was used to take photomicrographs. Histopathology severity was evaluated using the histopathology scoring system presented in Table 3.

Table 3: Histopathology Severity Grading System

Overall Severity Grading	Inflammation Scoring	Proportion of Airspace Area Affected
Inflammation was absent (within normal histologic limits)	—	None
Minimal-mild inflammation	—/+	≤10%
Mild to moderate inflammation	+	≥10 to <25%
Moderate to severe inflammation	++	≥25 to <50%
Severe inflammation	+++	≥50%

Note: Histopathology severity grading was based on the severity scoring of neutrophil infiltrate and the proportion of airspace area affected. The level of severity was judged from – to +++, which represents from none to severe. The proportion of airspace area affected was the ratio of the airspace area affected to the total airspace area of one view.

2.4.9. Immunohistochemistry

Staining of hamster lung sections was achieved on a (b) (4) used per the manufacturer's protocol. Tissue sections were dewaxed with (b) (4) at 72°C for 30 minutes and then subsequently rehydrated with graded alcohol washes and 1× Immuno Wash (b) (4). Heat-induced epitope retrieval was performed using (b) (4) heated to 100°C for 20 minutes. A peroxide block (b) (4) was applied for 5 minutes to quench endogenous peroxidase activity prior to applying the SARS-CoV-2 antibody ((b) (4)) at a dilution of 1:2000. Antibodies were diluted in Background Reducing Antibody Diluent (b) (4). The tissue was subsequently incubated with an anti-rabbit horseradish peroxidase polymer (b) (4) and colorized with 3,3'-diaminobenzidine chromogen for 10 minutes. The slides were counterstained with hematoxylin.

Sections were examined by a board-certified veterinary pathologist using an (b) (4) light microscope. An (b) (4) camera was used to take photomicrographs.

2.5. Data Presentation and Analysis

Geometric means or means are represented by the heights of bars or symbols, and error bars represent the corresponding SD. The LOD for each assay was the starting serum dilution; for example, if serum was diluted 1:25, then the LOD was 25. Dotted lines in figures indicate assay LODs. Two-way analysis of variance with Tukey corrections were performed for multiple comparisons between vaccine groups and significance between pre- and post- infection antibody titers was measured by multiple t tests with Holm-Sidak's correction for multiple comparisons (* $p \leq .05$, ** $p \leq .01$, *** $p \leq .001$, **** $p \leq .0001$; GraphPad Prism software version 6).

3. RESULTS

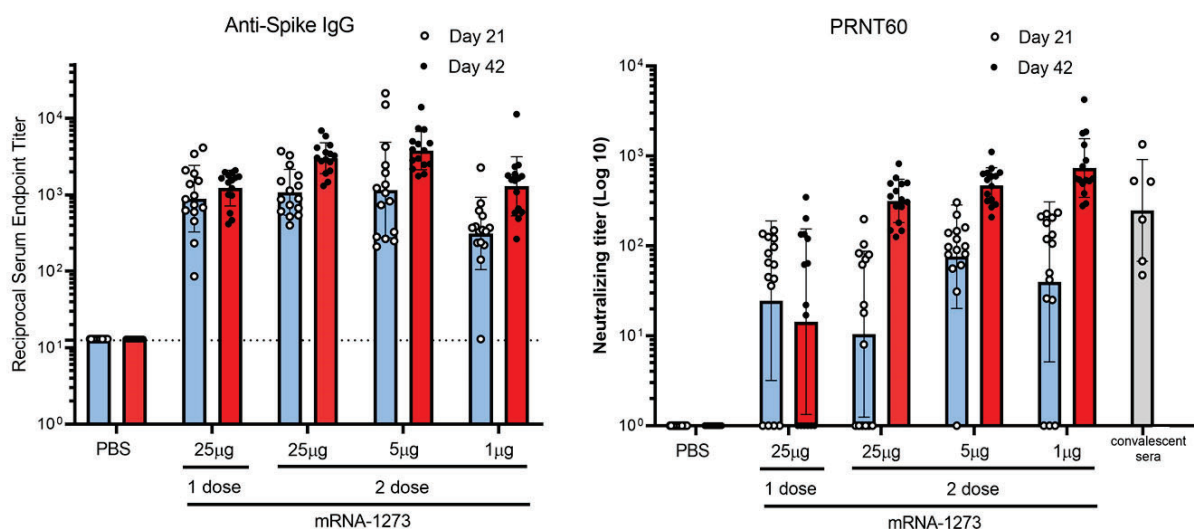
3.1. Spike-Specific IgG Antibody Responses and Neutralizing Activity

mRNA-1273 elicited S- and RBD-specific IgG antibody titers in all groups and neutralizing antibody responses in all animals in the prime/boost groups and a subset of animals in the prime-only group (Figure 1). S-specific and neutralizing antibody responses in post-boost samples were higher than those in post-prime samples in animals immunized with 1, 5, and 25 μ g of mRNA-1273 on a prime/boost schedule; seroconversion was only achieved after administration of the boost dose in these animals.

The neutralizing antibody titers were similar among all of the prime/boost groups. Post-boost (Day 42) neutralizing antibody titers were higher in animals in the prime/boost groups than the titers at Day 42 in animals immunized with a prime-only dose of 25 μ g of mRNA-1273.

The post-boost neutralizing antibody titers in animals immunized with 1, 5, and 25 μ g of mRNA-1273 on a prime/boost schedule met or exceeded the highest titers observed in convalescent sera.

Figure 1: Spike-Specific IgG Antibody Response and Neutralizing Antibody Response After mRNA-1273 Vaccination



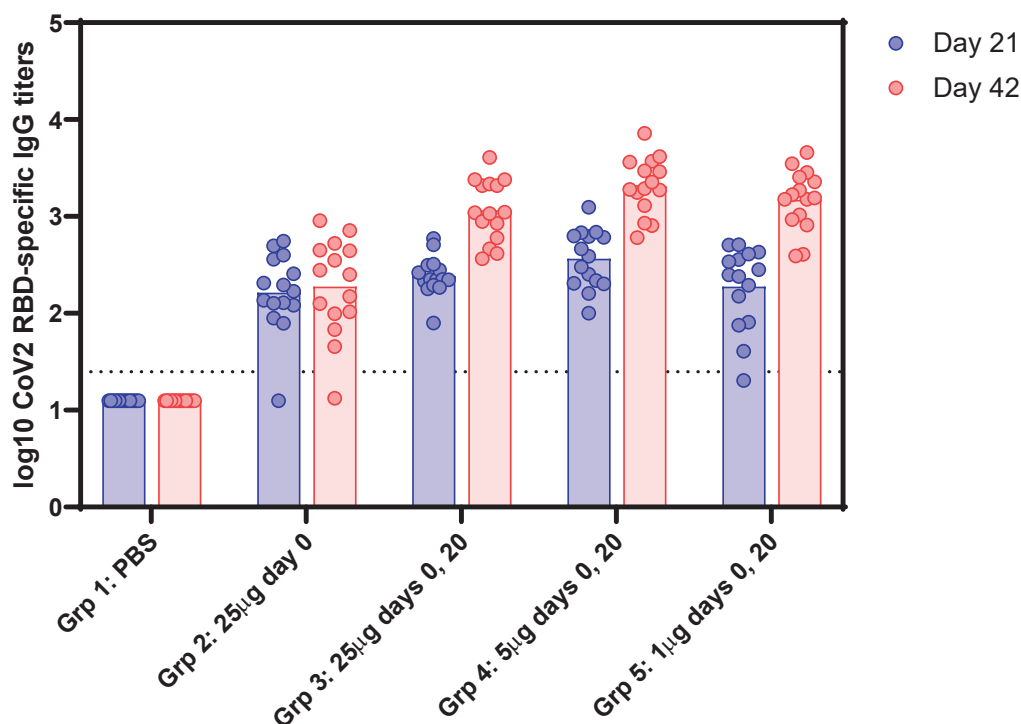
Abbreviations: IgG = immunoglobulin G; PBS = phosphate-buffered saline; PRNT60 = titer required to reduce viral plaques by 60%.

3.2. Receptor Binding Domain-Specific IgG Antibody Responses

mRNA-1273 elicited a robust RBD-specific IgG antibody response (Figure 2). The post-boost RBD-specific IgG antibody titers were higher than the post-prime titers in animals administered 2 doses of 1, 5, and 25 μ g of mRNA-1273. The RBD-specific IgG antibody titers were similar among animals administered 2 doses of 1, 5, and 25 μ g of mRNA-1273. The post-boost (Day 42)

RBD-specific IgG antibody titers were higher in animals administered 2 doses of 1, 5, and 25 μ g of mRNA-1273 than the titers at Day 42 in animals administered a single dose of 25 μ g of mRNA-1273.

Figure 2: Receptor Binding Domain-Specific IgG Antibody Responses After mRNA-1273 Vaccination



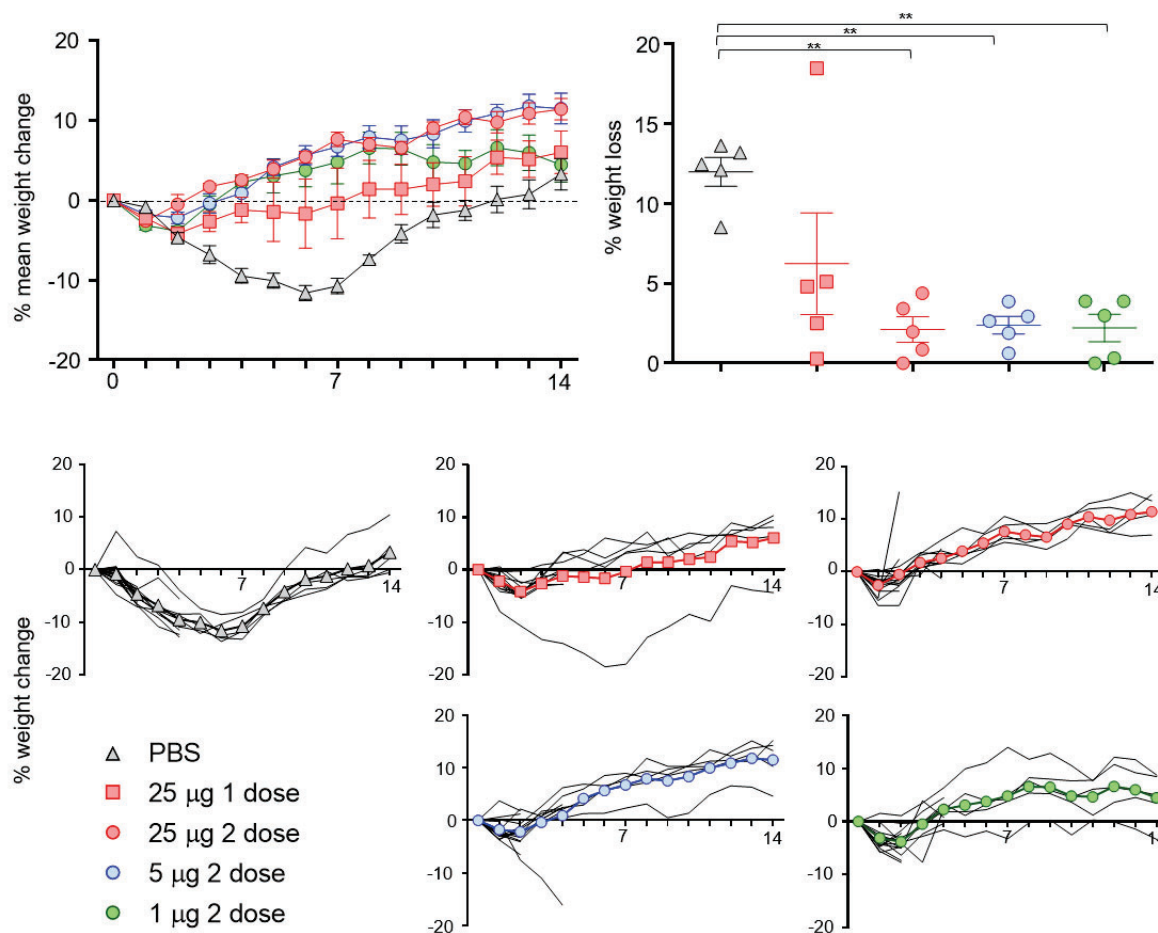
Abbreviations: CoV2 = SARS-CoV-2; Grp = group; IgG = immunoglobulin G; PBS = phosphate-buffered saline; RBD = receptor binding domain; SARS-CoV-2 = 2019 novel coronavirus.

3.3. Summary of Weight Change After SARS-CoV-2 Challenge

After challenge with wild-type SARS-CoV-2, minimal weight loss was observed in animals immunized with a prime/boost dose of 1, 5, and 25 μ g of mRNA-1273 through Day 14 post-challenge, except for one hamster in the 5 μ g mRNA-1273 group (Figure 3). Excluding the animals necropsied on Day 2 and Day 4 post-challenge, the mean maximum weight loss in the mRNA-1273 prime/boost groups was 2.25% over 14 days post-challenge. All but one animal immunized with a prime-only dose of 25 μ g of mRNA-1273 experienced little or no weight loss; the mean maximum weight loss in this group was 6.2%. In general, the mean weight in animals administered a prime-only dose of 25 μ g of mRNA-1273 remained lower than the mean weight of animals in the mRNA-1273 prime/boost groups through Day 14 post-challenge. In contrast, animals in the PBS group experienced marked weight loss by Day 6 post-challenge; the mean

weight increased from Day 7 to Day 14 post-challenge but remained lower in the PBS group than in all other groups. The mean maximum weight loss in the PBS group was 12%.

Figure 3: Summary of Percent Body Weight Changes



Abbreviation: PBS = phosphate-buffered saline.

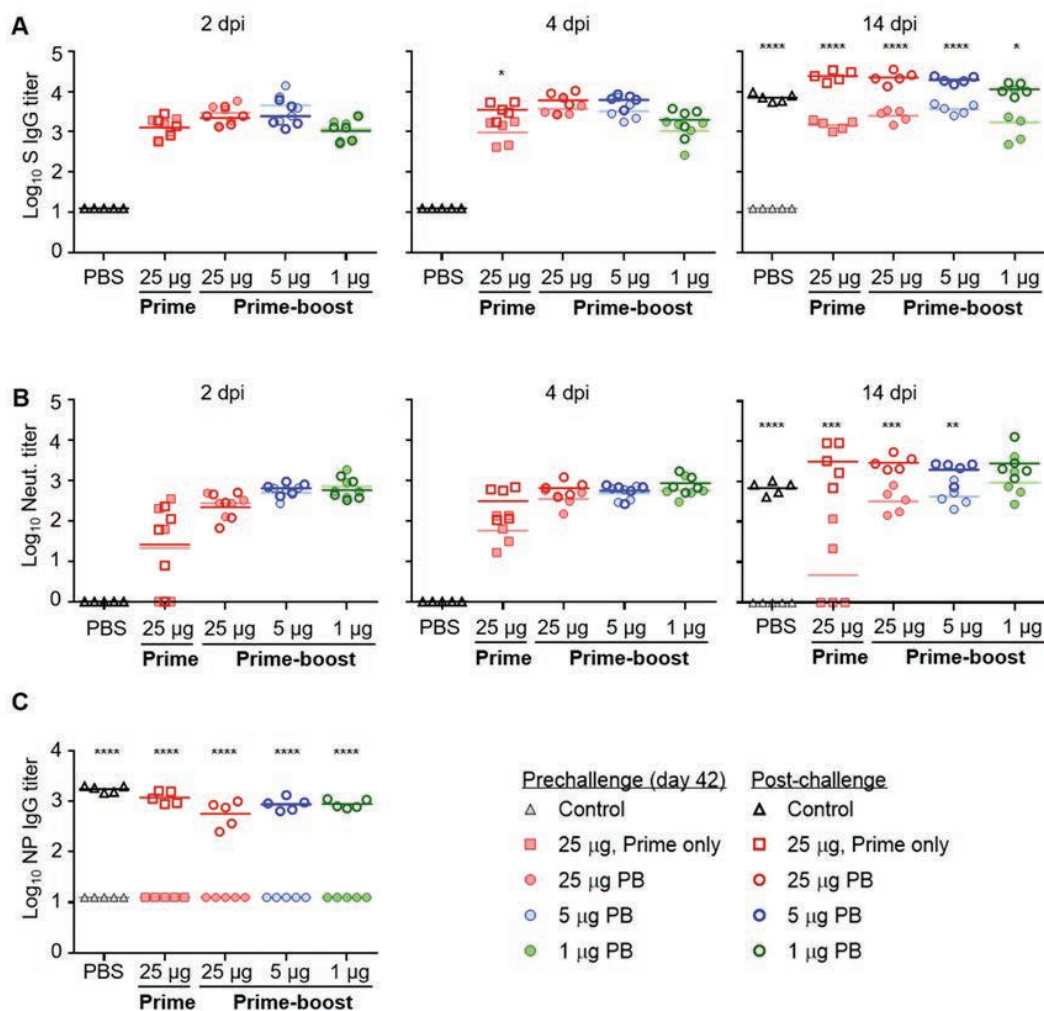
Note: ** $p \leq .01$.

3.4. Post-challenge Spike- and Nucleoprotein-Specific IgG Antibody Responses and Neutralization Responses

In the PBS group, S- and NP-specific IgG antibody titers and neutralizing antibody titers were detected at Day 14 post-challenge and were significantly higher than the titers at Day 42 pre-challenge (Figure 4A-C). In animals immunized with mRNA-1273 (all groups), S-specific IgG antibody titers and neutralizing antibody titers were detected prior to the virus challenge and the titers increased significantly by Day 14 post-challenge, except for neutralizing antibody titers in animals immunized with a prime/boost dose of 1 µg of mRNA-1273, in which the difference was not significant. In addition, NP-specific antibody titers were detected in all groups at Day 14 post-challenge, although the titers were lower in the mRNA-1273 groups than in the PBS group.

These results indicate that although mRNA-1273 protects Syrian hamsters from weight loss after SARS-CoV-2 challenge, this protection is likely not fully sterilizing as vaccinated animals mounted an anamnestic immune response against the virus challenge.

Figure 4: Spike- and Nucleoprotein-Specific IgG Antibody Response and Neutralizing Antibody Response Post-challenge



Abbreviations: dpi = days post-infection; IgG = immunoglobulin G; Neut = neutralizing; NP = nucleoprotein; PB = prime/boost; PBS = phosphate-buffered saline; S = spike.

Note: A) S-specific IgG titers before challenge (Day 42, solid symbols) or Day 2, Day 4, and Day 14 post-challenge (open symbols). B) Neutralizing antibody titers before challenge (Day 42) or Day 2, Day 4, and Day 14 post-challenge. C) NP-specific IgG antibody responses post-challenge.

Significance between pre- and post- infection antibody titers was measured by multiple t tests with Holm Sidak's correction for multiple comparisons; * $p < .05$, ** $p < .01$, *** $p < .001$, **** $p < .0001$.

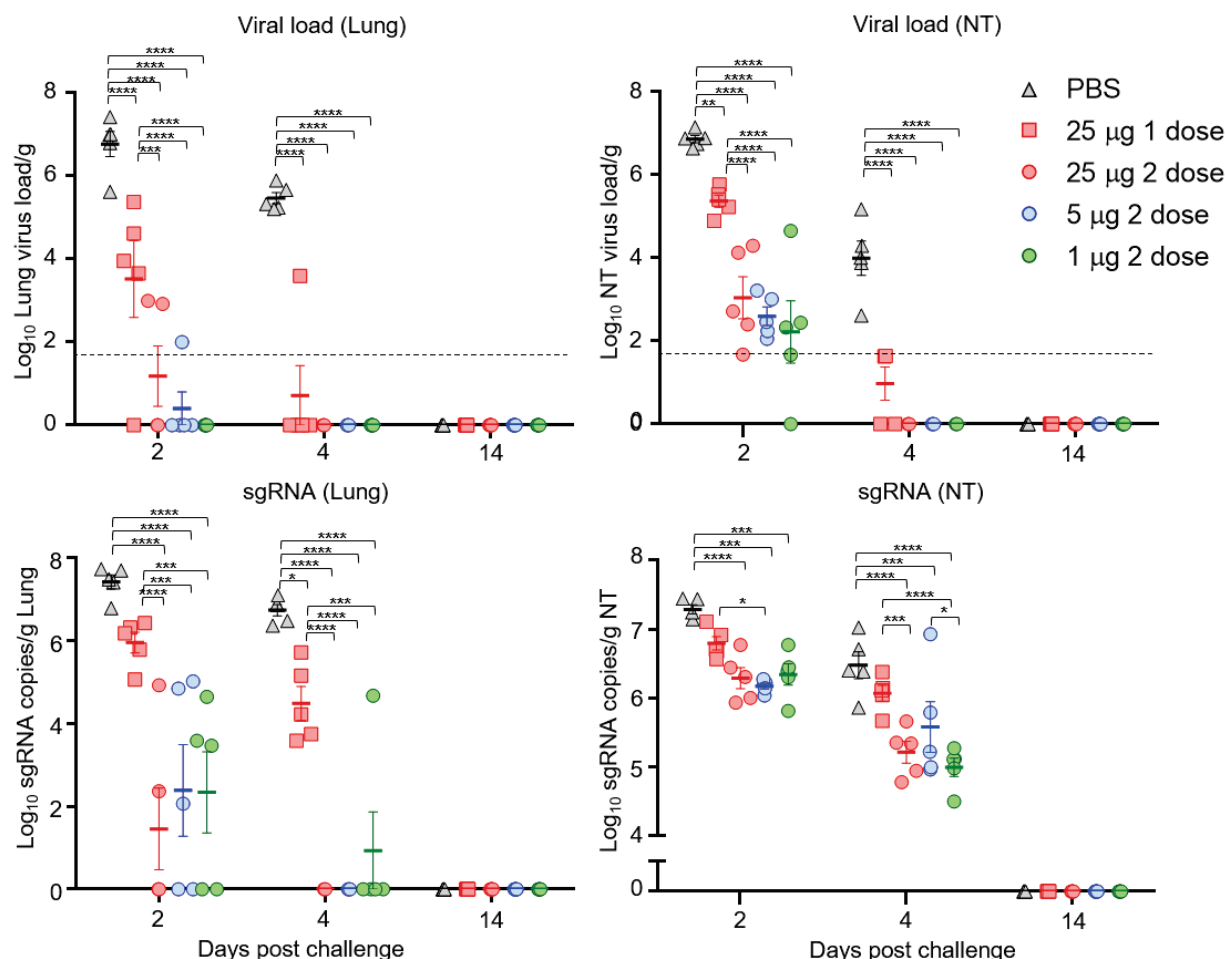
3.5. Viral Titers and sgRNA in Lungs and Nasal Turbinates

No protection from SARS-CoV-2 was observed in animals administered PBS, as substantial viral load (viral titers by plaque assay) and viral replication (sgRNA by qRT-PCR) were observed in the lungs and nasal turbinates on Day 2 and Day 4 post-challenge ([Figure 5](#)).

Measurement of viral load showed that animals immunized with 1, 5, and 25 µg of mRNA-1273 on a prime/boost schedule were protected from virus challenge, with mean viral titers that were below the LOD in the lungs and nasal turbinates at Day 4 post-challenge. In addition, all animals administered a prime/boost dose of 5 and 25 µg of mRNA-1273 and all but one animal administered a prime/boost dose of 1 µg of mRNA-1273 had no detectable viral replication (sgRNA) in the lungs by Day 4 post-challenge. In the nasal turbinates, all animals in the prime/boost groups were partially protected, as the viral replication (sgRNA) observed on Day 2 and Day 4 post-challenge was significantly lower than that in the PBS control group.

A prime-only dose of 25 µg of mRNA-1273 partially protected animals, as live virus and viral replication were observed on Day 2 post-challenge in lungs and nasal turbinates; however, the mean viral titers were lower than those in the PBS group. By Day 4 post-challenge, the mean viral titers in animals administered a prime-only dose of 25 µg of mRNA-1273 were below the LOD in lungs and nasal turbinates.

In all Syrian hamsters, including those in the PBS group, the virus cleared by Day 14 post-challenge.

Figure 5: Viral Load in Lungs and Nasal Turbinates

Abbreviations: NT = nasal turbinates; PBS = phosphate-buffered saline; sgRNA = subgenomic RNA.

3.6. Lung Histopathology

The lungs of hamsters were evaluated histologically at Day 2 (n = 5/group), Day 4 (n = 5/group), and Day 14 (n = 5/group) after challenge with wild-type SARS-CoV-2 (Table 4).

A mock-infected naïve group was included as a control and presented with moderately prominent alveolar interstitial hypercellularity after administration of media to mimic virus inoculum.

At Day 2 post-challenge, lung sections of animals in the PBS control group showed mild interstitial inflammation in the lungs, with some samples exhibiting a largely polymorphonuclear cellular infiltrate, consisting predominantly of neutrophils/heterophils, in and around the lung airways. By Day 4 post-challenge, inflammation was largely associated with perivascular and peribronchiolar regions in a focally diffuse or multifocal distribution affecting, on average, 30% to 50% of the lung, and this inflammation persisted through Day 14.

Over the 14-day infection course, lung sections of animals immunized with a prime/boost dose of mRNA-1273 displayed mild to moderate inflammation mostly involving the pulmonary interstitium; these interstitial changes were characterized by expanded alveolar capillary profiles, diminished alveolar spaces, and an influx of predominantly mononuclear cells. In the 1 µg prime/boost group, the predominant inflammatory phenotype over the 14-day infection course was mild to moderate interstitial inflammation, with one animal exhibiting perivascular-peribronchiolar inflammation at Day 4 post-challenge. However, one animal in the 1 µg prime/boost group exhibited a more severe pulmonary inflammatory response at Day 2 post-challenge that was characterized by mild to moderate edema and rare foci of hemorrhage and vascular congestion. At Day 4 post-challenge, lung sections of animals immunized with a prime-only dose of 25 µg of mRNA-1273 exhibited inflammation that was largely associated with perivascular and peribronchiolar regions in both a focally diffuse or multifocal distribution; however, this inflammation was generally milder than that in animals in the PBS control group at the same time point and did not persist through Day 14. One hamster in the 25 µg prime-only group presented with mild pulmonary edema and scant fibrin deposition at Day 4 post-challenge; this was not observed in the rest of the group. Regardless of vaccination schedule and dose, the airway lumen remained clear of inflammatory cells and cellular debris.

Lung sections were also examined for SARS-CoV-2 antigen by IHC at Day 4 post-challenge ([Figure 6](#)). Large amounts of viral antigen were found throughout the lungs of hamsters in the PBS control group. The presence of viral antigen was lower in animals immunized on a prime/boost schedule with mRNA-1273 than in animals on a prime-only schedule, and SARS-CoV-2 antigen was minimal to absent in animals immunized prime/boost with 5 and 25 µg of mRNA-1273. However, the 2 animals described above (one in the 1 µg prime/boost group and one in the 25 µg prime-only group) that exhibited the more severe histopathological phenotypes at Day 4 post-challenge had the highest levels of viral antigen among all vaccinated hamsters. Notably, these 2 animals did not exhibit drastic weight loss and the animal in the 25 µg prime-only group had live virus and virus replication in the lungs and nasal turbinates that was similar to that in other animals in their dose group (the animal in the 1 µg prime/boost group had discernably higher levels of viral replication [sgRNA] in the lungs than other animals in prime/boost groups). Interestingly, these 2 animals did not produce neutralizing titers after prime vaccination, which may emphasize the importance of vaccine dose and the evolving immune response after administration of a second dose.

Table 4: Assessment of Lung Inflammation and Viral Antigen Following SARS-CoV-2 Challenge of mRNA-1273 Immunized Hamsters

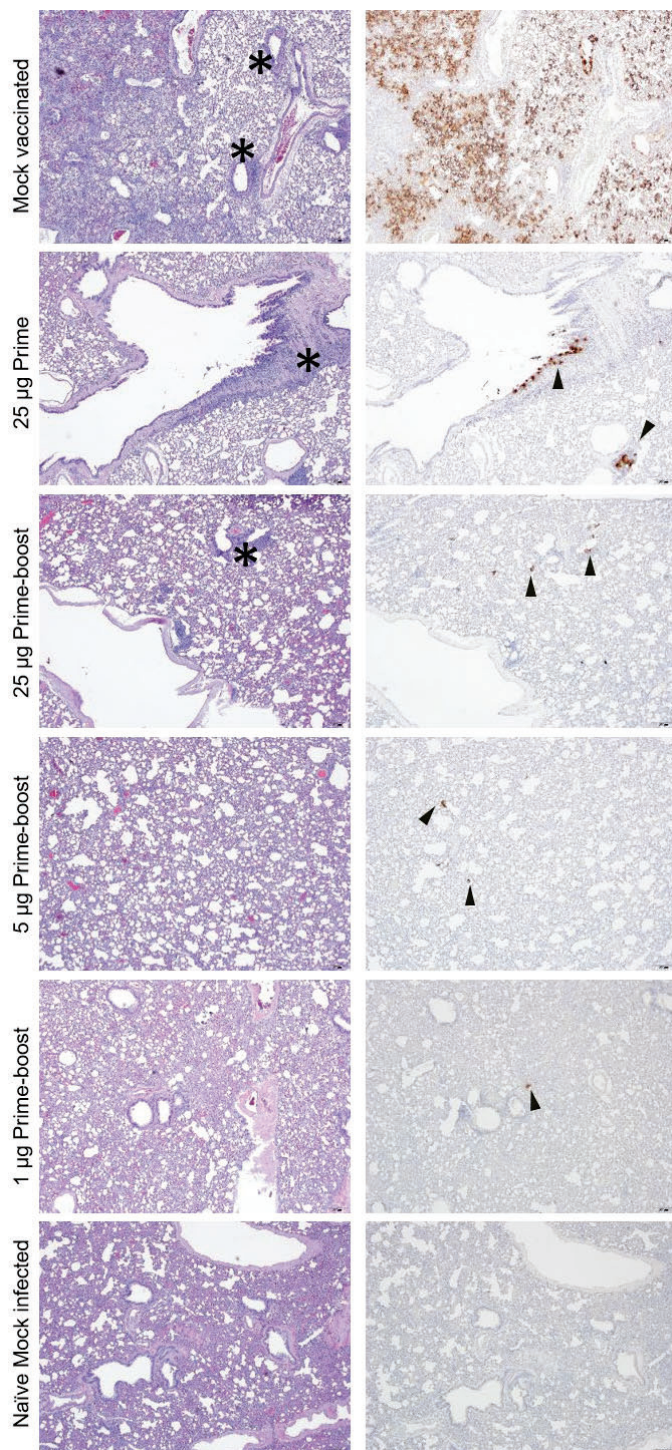
Dose Group	Inflammation (H&E) ^a			SARS-CoV-2 antigen (IHC) ^b
	Day 2 Post-challenge	Day 4 Post-challenge	Day 14 Post-challenge	Day 4 Post-challenge
PBS (prime-only)	+/- i	+ pv/pbr F	+ pv/pbr	+++
	+/- i	++ pv/pbr F	++ pv/pbr, i	+++
	+ i, aw	++ pv/pbr F	+ pv/pbr	++
	++ i, aw	+++ pv/pbr F	++ pv/pbr, i	++
	+ i, aw	+++ pv/pbr F	+ pv/pbr	+++
25 µg mRNA-1273 (prime-only)	++ i	+ pv/pbr F	+/- i	+
	+/- i	+ pv/pbr F	+/- i	+/- to +
	+/- i	+ pv/pbr F	+/- i	+
	+ i	+ pv/pbr F	+/- i	+
	+i	++ pv/pbr F, edema and fibrin	+/- i	++
25 µg mRNA-1273 (prime/boost)	+/- i	+ i	++ i	–
	+/- i	+ i	+ i	–
	+/- i	+ i	++ i	+/-
	++ i	+/- i	+ i	+/-
	+ i	+ i	++ i	–
5 µg mRNA-1273 (prime/boost)	+ i	+ i	+ i	+/-
	++ i	+/- i	+ i	–
	+/- i	+/- i	+/- i	+/-
	+ i	+/- i	++ i	–
	–	+ i	++ i	+/-
1 µg mRNA-1273 (prime/boost)	+ i	+ i	+ i	+/- to –
	++ i	+/- i	++ i	+/- to –
	+ i	++ pv/pbr F/i	+ i	++
	+/- i	+/- i	+ i	–
	+++ i, edema, rare hemorrhage	+i	+i	+/- to –
Naïve (mock infected)	NA	+ i	NA	–
	NA	+++ i	NA	–
	NA	++ i	NA	–
	NA	++ i	NA	–

Abbreviations: aw = airway; F = focal/multifocal; H&E = hematoxylin & eosin; i = interstitial; IHC = immunohistochemistry; NA = not applicable; PBS = phosphate-buffered saline; pv/pbr = perivascular/peribronchiolar; SARS-CoV-2 = 2019 novel coronavirus.

^a For inflammation scoring, – = minimal to absent inflammation; +/- = <10% tissue affected; + = >10% to <25% tissue affected; ++ = >26% to <50% tissue affected; +++ = >50% tissue affected.

^b For SARS-CoV-2 viral antigen scoring, – = minimal to absent; +/- = minimally abundant (but clearly present); + = mildly abundant; ++ = moderately abundant; +++ = abundant.

Figure 6: Histopathology and Immunohistochemistry of Lung Samples at Day 4 Post-challenge



Note: The first column shows hematoxylin and eosin staining of lung section to show severity of inflammation and the * reflects areas of perivascular/peribronchiolar inflammation (mostly mononuclear). The second column shows immunohistochemical staining (representative photomicrographs with original magnification $\times 4$ [scale bars, 200 μm]) with virus antigen (arrowhead) in lungs. "Mock vaccinated" represents the phosphate-buffered saline group

4. CONCLUSIONS

mRNA-1273 is immunogenic in Syrian hamsters. Immunization with mRNA-1273 elicited S- and RBD-specific IgG antibody titers in all groups and neutralizing antibody responses in all animals in the prime/boost groups and a subset of animals in the prime-only group. In general, the post-boost S- and RBD-specific IgG binding antibody titers and neutralizing antibody titers were higher than post-prime titers in animals immunized with 1, 5, and 25 µg of mRNA-1273 on a prime/boost schedule. Complete neutralizing seroconversion in all animals was only achieved after administration of the boost dose. Post-boost (Day 42) RBD-specific IgG binding antibody titers and neutralizing antibody titers were also higher in animals in the prime/boost groups than titers at Day 42 in animals immunized with a prime-only dose of 25 µg of mRNA-1273. The post-boost neutralizing antibody titers in animals immunized with 1, 5, and 25 µg of mRNA-1273 on a post/boost schedule met or exceeded the highest titers observed in convalescent sera.

Overall, minimal weight loss (mean maximum weight loss of 2.25%) was observed in animals immunized with a prime/boost dose of 1, 5, and 25 µg of mRNA-1273 through Day 14 post-challenge. Animals immunized with a prime-only dose of 25 µg of mRNA-1273 experienced little to no weight loss (mean maximum weight loss of 6.2%). In contrast, the PBS group showed marked weight loss by Day 6 post-challenge, and the mean weight remained lower in the PBS group than in all other groups (mean maximum weight loss of 12%).

In the PBS group, S- and NP-specific IgG antibody titers and neutralizing antibody titers were detected at Day 14 post-challenge, and the levels were significantly higher than the titers at Day 42 pre-challenge. In animals immunized with mRNA-1273 (all groups), S-specific IgG antibody titers and neutralizing antibody titers were detected prior to the virus challenge and the titers increased significantly by Day 14 post-challenge, except for neutralizing antibody titers in animals immunized with a prime/boost dose of 1 µg of mRNA-1273, in which the difference was not significant. In addition, NP-specific antibody titers were detected in all groups at Day 14 post-challenge, although the titers were lower in the mRNA-1273 groups than in the PBS group. These results indicate that although mRNA-1273 protects Syrian hamsters from weight loss after SARS-CoV-2 challenge, this protection is likely not fully sterilizing as vaccinated animals mounted an anamnestic immune response against the virus challenge.

Measurement of viral load showed that animals immunized with 1, 5, and 25 µg of mRNA-1273 on a prime/boost schedule were protected from virus challenge, with mean viral titers that were below the LOD in the lungs and nasal turbinates at Day 4 post-challenge. In addition, all animals administered a prime/boost dose of 5 and 25 µg of mRNA-1273 and all but one animal administered a prime/boost dose of 1 µg of mRNA-1273 had no detectable viral replication (sgRNA) in the lungs by Day 4 post-challenge. In the nasal turbinates, all animals in the prime/boost groups were partially protected, as the viral replication (sgRNA) observed on Day 2

and Day 4 post-challenge was significantly lower than that in the PBS control group. A prime-only dose of 25 µg of mRNA-1273 partially protected animals, as live virus and viral replication were observed on Day 2 post-challenge in lungs and nasal turbinates; however, the mean viral titers were lower than those in the PBS group, and by Day 4 post-challenge, the mean viral titers were below the LOD.

Lungs of hamsters evaluated histologically after challenge with wild-type SARS-CoV-2 showed that over the 14-day infection course, animals immunized with a prime/boost dose of mRNA-1273 generally displayed mild to moderate inflammation with viral antigens that were either absent or minimally abundant at Day 4 post-challenge. Lung sections of animals immunized with a prime-only dose of 25 µg of mRNA-1273 exhibited inflammation that was largely associated with perivascular and peribronchiolar regions in both a focally diffuse or multifocal distribution, but this inflammation was generally milder than that in animals in the PBS control group at the same time point and did not persist through Day 14. One outlier animal in each of the 1 µg prime/boost and 25 µg prime-only groups exhibited more severe histopathological phenotypes, with the highest levels of viral antigen detected among all vaccinated hamsters. These 2 animals did not exhibit drastic weight loss and the animal in the 25 µg prime-only group had live virus and virus replication in the lungs and nasal turbinates that was similar to that in other animals in their dose group.

Overall, mRNA-1273 administered on a prime/boost dosing schedule is immunogenic and efficacious and provides protection from SARS-CoV-2 in Syrian hamsters.

5. REFERENCES

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6. REPORT APPROVAL

Study Director:

(b) (6)

(b) (6)

Date

UTMB
301 University Blvd
Galveston, TX 77555.

7. APPENDICES

APPENDIX 1 SUMMARY OF ANALYSIS: LOT AMPDP-200005 RELEASE



200 Tech Square • Cambridge, MA 02139
phone 617-714-6500 • fax 617-583-1998

SUMMARY OF ANALYSIS

Product Description: mRNA-1273 Drug Product

Project Code: mRNA-1273 **Batch ID:** B-20200201-01404

Manufacturer: Moderna **Sample ID:** S-200201-00010

Moderna Lot Number: AMPDP-200005 **Date of Manufacture:** 01-Feb-2020

mRNA Lot Number: (b) (4) **Expiry Date:** N/A

Volume: 0.6 mL **Shipping Temperature:** -70°C

Storage Temperature: -70°C

Final Buffer 10.7 mM Sodium Acetate; 8.7% Sucrose; 20 mM Tris, pH 7.5

Container Closure: 2R vial, USP Type I; Traditional rubber stoppers are West 4432/50 (chlorobutyl rubber) composition with a Flurotec coating.

Test	Method	Target Attributes	Results
Identity	DPTM-0023	(b) (4)	(b) (4)
Appearance	DPTM-0030	White to off-white dispersion. May contain visible, white or translucent product-related particulates	Conforms, White to off-white dispersion; essentially free of visible particulates
RNA Content	DPTM-0024	(b) (4)	(b) (4)
RNA Purity	DPTM-0046	(b) (4)	(b) (4)
% Encapsulation	DPTM-0063	(b) (4)	(b) (4)
Filtered Mean Particle Size	DPTM-0012	(b) (4)	(b) (4)
Filtered Polydispersity Index		(b) (4)	(b) (4)



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Test	Method	Target Attributes	Results
Lipid Identification:			
SM-102	DPTM-0019	(b) (4)	
Cholesterol			
DSPC			
PEG2000-DMG			
Lipid Content (mg/mL)			
SM-102	DPTM-0019	(b) (4)	(b) (4)
Cholesterol			
DSPC			
PEG2000-DMG			
Lipid Impurities (% Area)			
Individual Impurities	DPTM-0019	(b) (4)	(b) (4)
Total Impurities			
pH	USP <791> DPTM-0028		
Osmolality	USP <785> DPTM-0029		
Bacterial Endotoxin	USP <85>		
Bioburden	USP<61>		
Particulate Matter	USP <788> Method 2		



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REVISION HISTORY

Revision #	Change Details	Author	Effective Date
1.0	Introduction of a New Document	(b) (6)	Date of Approval
2.0	Added in results for Identity and particulate matter	(b) (6)	Date of Approval
3.0	Added in results for bioburden testing and updated bacterial endotoxin results with values from ACCI (DPAD-00577)	(b) (6)	Date of Approval
4.0	Added correct test methods for RPIP and (b) (4)	(b) (6)	Date of Approval